

Reshaping Occupational DNA: Evolution Strategies in the Generative AI Era

Summary

When a generative model can draft, code, and summarize on demand, curricula built around information transfer become fragile; **what remains valuable is the structure of work—tasks, judgment, and responsibility**. We therefore model the Gen-AI transition from the ground up: occupations are decomposed into an auditable **task DNA** using O*NET (task importance and frequency), and Gen-AI impact is expressed on a **nine-dimension capability scale** that separates *substitution pressure* from *complementarity gains*. An adoption-driven dynamic then converts these task-level signals into interpretable occupational trajectories (which tasks shrink, which expand, and when the occupational boundary effectively changes).

To ensure our conclusions are not tailored to a single niche, we select three representative occupations by identifying an **“occupational triangle”** with maximum heterogeneity: *Robotics Engineers* (SOC 17-2199.08), *First-Line Supervisors of Mechanics, Installers, and Repairers* (SOC 49-1011.00), and *Court Reporters and Simultaneous Captioners* (SOC 27-3092.00). **The selection is stable under candidate-pool expansion** and yields three distinct futures: a robotics pathway dominated by complementarity and task reallocation; a skilled-trade supervisory pathway where routine documentation and scheduling are compressed while field judgment and coordination become more central; and a court-reporting pathway where core production is more directly exposed to substitution, forcing a stronger toolchain redesign.

We translate these occupational insights into institutional policy and curriculum action. For each occupation we recommend one program that can realistically institutionalize Gen-AI resources (MRSD at CMU; UTI Diesel; Generations College Court Reporting) and then evaluate each course using a **three-axis meta-model**: (i) marginal employability contribution based on O*NET task coverage, (ii) humanistic value spanning civic responsibility, agency, well-being, and lifelong learning, and (iii) AI-overlap v2, defined by each course’s nine-dimension footprint weighted by the previously estimated Gen-AI capability strengths. The resulting **KEEP/-TRANSFORM/PRUNE** decisions are designed for portability (course title + description suffices) and emphasize **transforming high-overlap courses into human-AI collaboration**, evidence-based reasoning, and ethical application rather than indiscriminate deletion.

Finally, we test the most consequential uncertainty—the definition of the target occupation set—using narrow/medium/wide variants. Across all three programs, **course-level decisions and the set of courses flagged for transformation remain unchanged** (label stability and transformation-set agreement are both 1.00), while fine-grained employability rankings can vary, establishing a clear generalization boundary: **the framework transfers across institutions, but the intended employment target must be explicitly re-specified when extrapolating**.

Keywords: Generative AI; Task DNA; Employability alignment; Humanistic

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1 Introduction

1.1 Background and Motivation

Generative Artificial Intelligence (Gen-AI) is evolving from a niche auxiliary tool into an indispensable foundational resource across all industries. With its rapidly growing capabilities, Gen-AI is not only optimizing workflows but fundamentally reshaping the task structure of occupations. Correspondingly, higher education must synchronously adjust its training systems to respond to the real-world challenge of “skill structures being rewritten by technology” [19, 37].

1.2 Problem Restatement

The core question posed by the problem is singular: how can post-secondary educational institutions better integrate Gen-AI into their current training programs? This core question decomposes into the following workflow:

First, we need to select one specific occupation from each of the three categories—**STEM, Trade, and Arts**—and characterize the future changes of these occupations under the influence of Gen-AI. Here, “future” can be understood as requiring a defined prediction window, or viewed as a broad time horizon. “Current trajectory” can be interpreted as the penetration and adoption rate of Gen-AI in the respective industry; “expected impact” refers to the measurable changes this penetration brings to job structures and practitioners’ working methods.

Second, after completing the occupation-level analysis, the problem further requires: selecting a “corresponding specific institution and specific program” for each occupation, and translating recommendations into concrete improvement measures for these three programs. The requirement is to use mathematical model outputs to provide evidence-based guidance rather than remaining at the level of principled slogans. This aligns with recent higher education research conclusions that institutions need to institutionalize GenAI usage boundaries, learning outcomes, and assessment methods together, rather than merely issuing generalized statements [10, 13].

Finally, beyond the recommendations given in the problem statement, the team must independently consider additional evaluation dimensions and explain whether the model can be generalized to other institutions or programs, explicitly stating the conditions and boundaries for generalizability.

2 Assumptions and Notation

2.1 Fundamental Assumptions

To ensure the reproducibility and auditability of our model, we formalize the core premises of the modeling process as follows:

- **Baseline Task Structure Stability Assumption.** *Justification:* When $A(t) = 0$, the task structure remains at baseline; that is, its endogenous changes are stationary. This simplification allows us to isolate the effect of GenAI adoption from other confounding factors in occupational evolution.
- **Monotonic Adoption Rate Assumption.** *Justification:* The current GenAI adoption rate increases monotonically from 0 to 1, characterizing the staged penetration during the technology diffusion process. Historical data on technology adoption (e.g., internet, mobile devices) consistently shows S-curve patterns with monotonic growth phases.

2.2 Notation

The global notation used in this paper is defined in the following table.

Table 1: Notation Definitions

Symbol	Description	Domain/Unit
i	Occupation index	$\mathbb{Z}^+$
j	Task index	$\mathbb{Z}^+$
t, t_0, t_1	Time and comparison time points	Years
x	Candidate occupation	—
$f(x)$	Occupation feature vector ($digit(x), physical(x), iprisk(x)$)	$[0, 1]^3$
w_{ij}	Task weight	$[0, 1]$
s_{ij}	Task substitution intensity	$\{0.10, \dots, 0.90\}$
c_{ij}	Task augmentation intensity	$\{0.10, \dots, 0.90\}$
μ_{ij}^{task}	Task net impact: $c_{ij} - s_{ij}$	$[-1, 1]$
$A(t)$	GenAI adoption rate trajectory	$[0, 1]$
$q_{ij}(t)$	Unnormalized task demand	$\mathbb{R}_{>0}$
r_{ij}	Structural change contribution	$[0, 1]$
$\text{rank}_i(j, t)$	Rank of task j in occupation i at time t	$\mathbb{Z}^+$
$I_i(t_0, t_1)$	Inversion count	$\mathbb{Z}_{\geq 0}$
$J_i(t_0, t_1; m)$	Top- m turnover rate	$[0, 1]$
N_i	Total number of tasks in occupation i	$\mathbb{Z}^+$
θ_I, θ_J	Inversion intensity threshold and turnover rate threshold	$(0, 1)$
k	Capability dimension index (D1–D9)	$\{1, \dots, 9\}$
M_k	Dimension representative score	$[0, 1]$
s_k, c_k	Dimension substitution/augmentation intensity	$[0, 1]$
ρ_k	Direction coefficient ($\rho_k = c_k - s_k$)	$[-1, 1]$

3 Model Construction and Implementation

3.1 Occupation Selection Across Three Domains

For occupation selection, we do not treat this as an assumption but rather as an optimization problem. We use the heterogeneity among the three occupations as the objective function, aiming to ensure the model has maximum explanatory validity. We employ “complete data support” and “policy relevance” as constraints, ultimately ensuring that subsequent models are built upon a reproducible, data-driven framework.

3.1.1 O*NET-Based Occupation Feature Parameters

First, we need to position the three occupations along three feature dimensions to establish mechanistic contrast among them. The three domains specified by the problem naturally possess such conditions. We derive the feature parameters for the three occupations as $digit(x)$, $physical(x)$, and $iprisk(x)$. We then score all U.S. occupations on these three dimensions. To avoid subjective assignment, we adopt the **O*NET** occupational database maintained by the U.S. Department of Labor [30] as the unified data source. O*NET, based on expert surveys and practitioner feedback, provides numerous *numerical scale descriptors* for each occupation, with most indicators naturally given as real numbers between 0 and 100, suitable for direct use in quantitative modeling.

We do not attempt to find a “uniquely correct” indicator for each dimension. For each feature axis, we select O*NET descriptors that semantically correspond directly to the three dimensions and are downloadable [30], using them as proxy variables for that axis. We directly exclude occupations with missing descriptor fields. Based on these constraints, we identify three dominant dimensions:

- **Digitization degree** $digit(x)$: The extent to which core tasks can be digitally represented, automated, or augmented;
- **Physical anchoring** $physical(x)$: The degree of dependence on on-site physical presence or manual interaction with the environment for task execution;

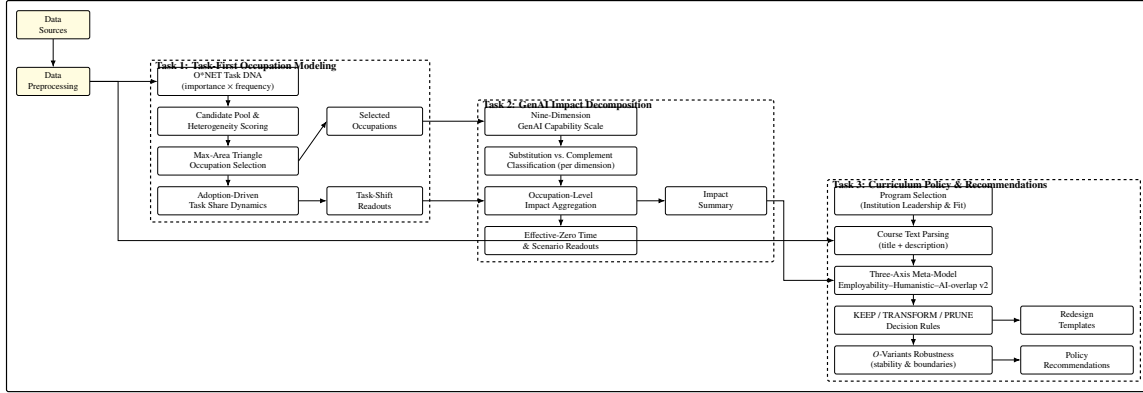


Figure 1: Overall Structure of Our Modeling Pipeline (Task 1–3).

- **Intellectual property and attribution risk $iprisk(x)$:** The sensitivity of value creation to authorship, originality, and ownership.

Table 2: Mapping Between Three-Dimensional Feature Axes and O*NET Descriptors

Feature Axis	Objective Measurement	O*NET Descriptor
$digit(x)$ Digitization	Dependence on computers/electronic media in work activities	Working with Computers [28]; E-Mail [23]
$physical(x)$ Physical Anchoring	Manual operation + physical posture demands	Using Hands to Handle Objects [27]; Kneeling/Crouching [24]
$iprisk(x)$ IP/Attribution Risk	Whether output centers on original expression/creative contribution	Originality [25]; Thinking Creatively [26]

For reproducibility and auditability, we maintain the access links and descriptor IDs for the six descriptors in Appendix Table 9.

Based on this, we filter all occupations that possess all six descriptor fields. For each occupation x , O*NET provides corresponding descriptor scores, so we let $v_m(x)$ denote its raw score under O*NET descriptor m (0–100). We apply min-max normalization across the entire occupation set:

$$z_m(x) = \frac{v_m(x) - \min_y v_m(y)}{\max_y v_m(y) - \min_y v_m(y)} \in [0, 1], \quad (1)$$

where y ranges over all occupations. Let the descriptor set for feature axis $k \in \{digit, physical, iprisk\}$ be D_k (see Table 2), and we take the equal-weight average of multiple descriptors under the same feature axis:

$$feature_k(x) = \frac{1}{|D_k|} \sum_{m \in D_k} z_m(x). \quad (2)$$

This yields the numerical values for $digit(x)$, $physical(x)$, and $iprisk(x)$.

Each candidate occupation x is represented by a normalized three-dimensional feature vector, serving as a

data point in a 3D scatter plot.

$$f(x) = (\text{digit}(x), \text{physical}(x), \text{iprisk}(x)) \in [0, 1]^3, \quad (3)$$

To select three occupations that constitute maximum heterogeneity (diversity) among qualifying occupations, we approach this geometrically: finding three data points that form the largest triangular plane in the three-dimensional coordinate system. Following the standard area formula:

$$\text{Area3D}(x_S, x_T, x_A) = \frac{1}{2} \|(f(x_T) - f(x_S)) \times (f(x_A) - f(x_S))\|_2, \quad (4)$$

where $x_S \in \text{STEM}$, $x_T \in \text{Trade}$, $x_A \in \text{Arts}$.

We then visualize using Python (Figure 2 and Figure 3).

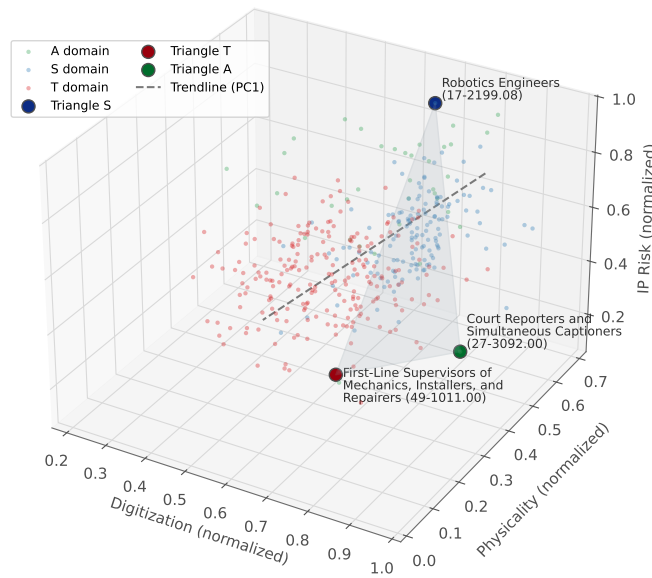


Figure 2: Three-dimensional heterogeneity point cloud of all occupations and optimal triangle plane (based on normalized O*NET descriptors)

We first solve for the unconstrained maximum-area triangle on the full point cloud, but the result included an occupation not officially classified under any of the three domains. Therefore, we add the constraint that selections must belong to O*NET-maintained target domains within the feasible region. The final constrained optimal three points are STEM (Robotics Engineers), Trade (First-Line Supervisors of Mechanics, Installers, and Repairers), and Arts/Media (Court Reporters and Simultaneous Captioners). The triangle formed by these three has an area of 0.262221.

As shown in Figure 2, the three-dimensional scatter plot intuitively displays the final optimization result. To illustrate the pairwise heterogeneity among occupations, we further plot the two-dimensional projections of this “occupational triangle” in Figure 3.

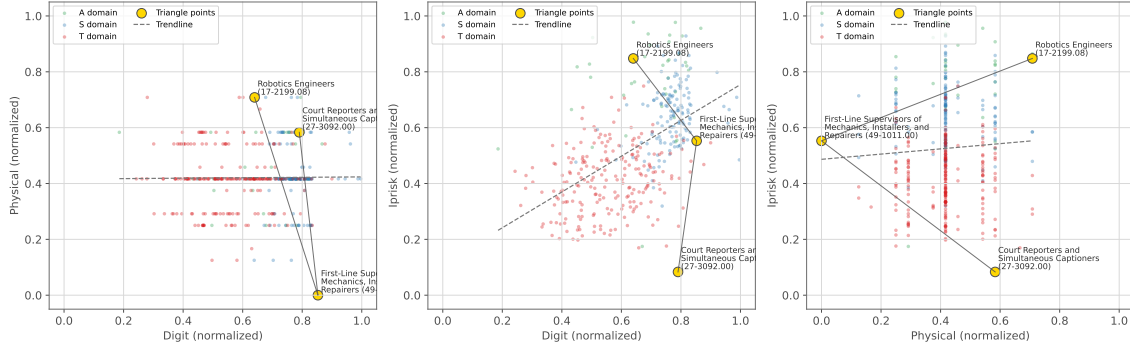


Figure 3: Two-dimensional projections of the occupation heterogeneity space with optimal three-point annotations (2D area formula in Appendix .2)

Analyzing the two-dimensional projections: Robotics Engineers have slightly lower digitization than Court Reporters and Simultaneous Captioners, and are located in the high physical/high IP risk region; First-Line Supervisors of Mechanics, Installers, and Repairers are in the low physical/medium IP risk region; while Court Reporters and Simultaneous Captioners are in the medium physical/low IP risk region. We observe that the three occupations maintain substantial differentiation in any two-dimensional combination, validating their selection as domain representatives. (The O*NET-SOC codes are: STEM: 17-2199.08; Trade: 49-1011.00; Arts: 27-3092.00 [29].)

3.2 Model Construction for Occupational Evolution

The second core requirement of the problem is to construct a *data-driven* future evolution model for the three selected occupations, emphasizing the combination of “current trajectory” and related driving factors. First, from first principles, an occupation is defined by its task composition (DNA); therefore, under Gen-AI’s influence, its future Task DNA will inevitably undergo structural “mutation.” Thus, attempting to predict future job demand for that occupation is logically problematic, as the occupation at that time will no longer be the same as today.

Therefore, we do not directly predict future occupational job demand, but instead approach from a more fundamental angle, studying the structural changes in occupational Task DNA under the current Gen-AI development trajectory.

For ease of understanding, we draw Figure 4 to illustrate the occupational evolution process.

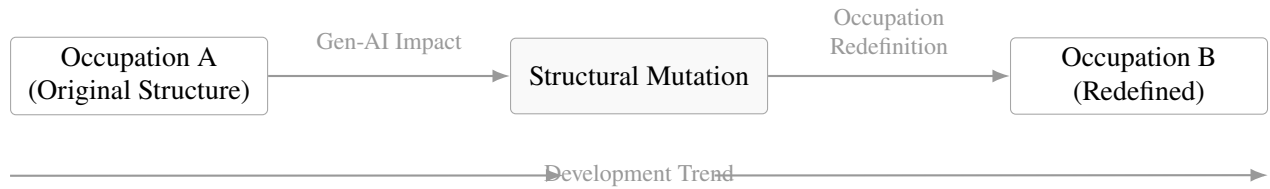


Figure 4: Deductive path from task structure rearrangement to occupation redefinition

3.2.1 Task-Level Dynamic Prediction Model

Since the official raw task description fields for occupations are too numerous, filtering is required. By constructing Task DNA task weights w_{ij} , combined with authoritative data sources, we calculate substitution/augmentation intensities (s_{ij}, c_{ij}) at the task (rather than occupation) level, obtaining the task-level net impact for the occupation as

$$\mu_{ij}^{task} = c_{ij} - s_{ij}, \quad (5)$$

and define “current trajectory” as society’s AI adoption rate $A(t) \in [0, 1]$, which drives the evolution of task demand over time.

Furthermore, we begin constructing the mathematical model for task-level prediction.

For occupation i , we select its **top N core tasks** as its Task DNA (task index $j = 1, \dots, 20$). Each task j is taken from O*NET's task description $Task_j$ for this occupation [30]. The database provides importance and frequency scoring data for each occupation's task composition, so we define two descriptors based on this data: **Task Importance** IM_j (range 0–100); **Task Frequency** FR_j (range 0–100, or equivalent frequency scale).

We define task weight as the normalized value of the product of importance IM_j and frequency FR_j . Let the unnormalized weight be:

$$\tilde{w}_j = IM_j \cdot FR_j, \quad (6)$$

then normalize:

$$w_{ij} = \frac{\tilde{w}_j}{\sum_{k=1}^{20} \tilde{w}_k}, \quad \sum_{j=1}^{20} w_j = 1. \quad (7)$$

Based on Equation (7), we compute each occupation's task weights using Python programming (if the selected SOC code lacks the task frequency scale FR_j , we use only importance values, i.e., $\tilde{w}_j = IM_j$), then select the Top Tasks (up to 20) for the three occupations and visualize them. Distribution results are shown in Figure 5; task text legends are in Appendix .3.

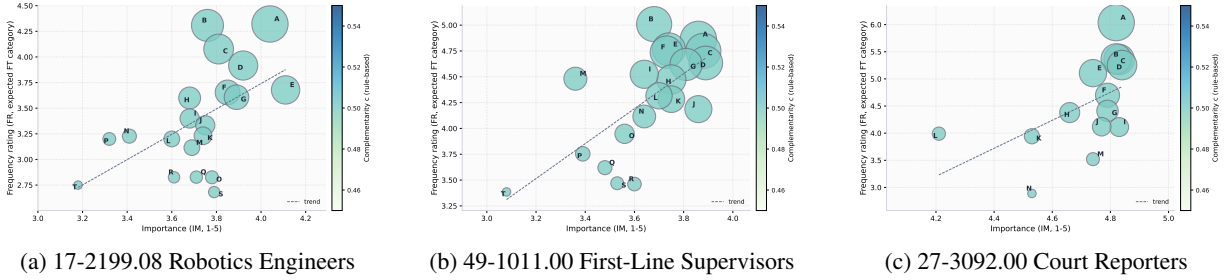


Figure 5: Task DNA weight distribution for three occupations (IM–FR scatter plot), point area proportional to w_j . Text legends for letters in Appendix .3.

Combining the Task DNA of the three occupation types, we provide targeted interpretations: for **Robotics Engineers**, occupational value is primarily driven by a few high-weight technical tasks; for **First-Line Supervisors (Mechanics/Installation/Repair)**, points are more dispersed with higher proportions of medium-weight tasks, reflecting the coexistence of management, coordination, and on-site judgment tasks, with structural change contributed by multiple tasks; while **Court Reporting and Captioning** highly depends on a few core language processing tasks.

After obtaining the Top Tasks for the three occupations, we need to map them to current Gen-AI capabilities to study the impact intensity on each Task. Thus, we similarly need to dissect current Gen-AI capabilities, dividing them into multiple capability dimensions. The following three points serve as the main basis for division:

- Coverage of major work activity types: from technical problem-solving to language processing, encompassing the main categories of occupational tasks
- Quantifiable and auditable: each dimension has corresponding authoritative benchmark tests, avoiding subjective evaluation
- Matching the task composition of the current three occupations: each Task DNA can be measured through publicly available AI test scores.

To avoid subjective continuous scoring, we construct a horizontal discrete capability scale for capability segmentation (M_k):

High (≥ 0.85) Moderately High (0.70–0.85) Medium (0.50–0.70) Low (0.30–0.50) Very Low (< 0.30)

with representative normalized scores M_k taken within each interval (specific mapping in Table 3).

We also introduce the direction coefficient (ρ_k) to characterize the bias strength of “substitution/complementarity/balance”: if $\rho_k < 0$: substitution-biased (AI can independently complete deliverables); if $\rho_k > 0$: complementarity-biased (AI provides suggestions/assistance, human must be in the loop); if $\rho_k = 0$: balanced (both strong, net effect uncertain). $\rho_k = c_k - s_k$ represents the direction of GenAI’s impact on that dimension: positive for complementarity (improving human efficiency increases demand), negative for substitution (automation decreases demand).

The final results are summarized in Table 3.

Table 3: GenAI Nine-Dimensional Capability Mapping Summary

Dimension (ID)	Representative Metrics (Source & Score)	M_k	Capability Level (Threshold Rules)	ρ_k	s_k	c_k
D1 Engineering & Technical Problem Solving	MMLU 86.4%; ARC 96.3% (GPT-4 Re-port[31])	0.90	High (≥ 0.85)	+0.35	0.50	0.85
D2 Data Analysis & Evaluation	GSM8K 92.0%; DROP 80.9 (GPT-4 Re-port[31])	0.80	Moderately High (0.70–0.85)	+0.20	0.30	0.50
D3 Algorithm Programming & Automation	HumanEval 67.0%; SWE-bench fix 20%[4, 34]	0.67	Medium (0.50–0.70)	−0.20	0.90	0.70
D4 On-Site Perception & Physical Operation	AutoGPT physical environment tasks 24%; Visual recognition $\approx 85\%$ [33, 5]	0.25	Low (0.30–0.50)	+0.40	0.10	0.50
D5 Planning & Workflow Scheduling	AutoGPT autonomous task completion rate $\sim 65\%$; Tool use success rate $\sim 80\%$ [33]	0.65	Medium (0.50–0.70)	−0.10	0.60	0.50
D6 Collaboration, Communication & Training	Human preference: GPT-4 win rate 70.2%[31]	0.70	Moderately High (0.70–0.85)	+0.45	0.30	0.75
D7 Supervision, Management & Quality Control	HellaSwag 95.3%[31]	0.80	Moderately High (0.70–0.85)	0.00	0.68	0.68
D8 Language Understanding & Text Generation	Un- Whisper WER ≈ 0.011 ; GPT-4 text preference win rate 70%[32, 31]	0.95	High (≥ 0.85)	−0.40	0.90	0.50
D9 Information Management & Retrieval	DocVQA 99.16%; Text-to-SQL $> 80\%$ [15, 38]	0.90	High (≥ 0.85)	0.00	0.30	0.30

Note: D9’s DocVQA 99.16% can be viewed as upper-bound evidence for document QA scenarios, not equivalent to general capability across all retrieval contexts and database scales. D4/D5 use AutoGPT end-to-end completion rate and tool use success rate as lower-bound evidence to avoid overstating the automation potential of “on-site operations/workflow scheduling.”

The table above only shows GenAI’s nine dimensions, while Figure 6 provides a more intuitive visualization of each dimension mapping to occupational task composition.

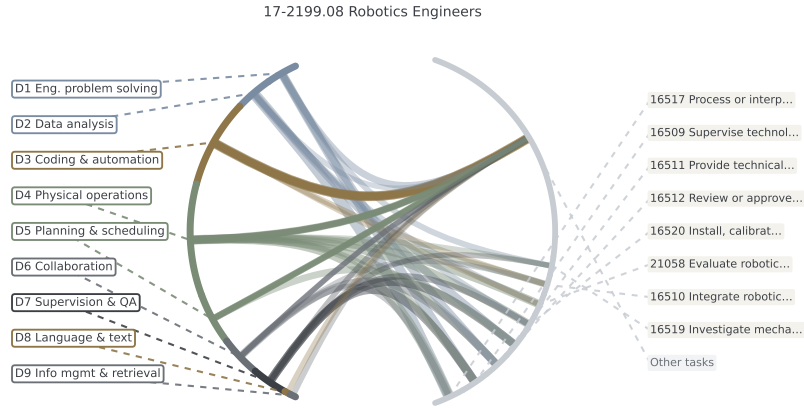


Figure 6: GenAI Capability → Task DNA mapping chord diagram (Robotics Engineers, 17-2199.08).

Overall, GenAI demonstrates high capability in language and coding dimensions, exhibiting substitution trends, while in dimensions involving physical operations and management, it still primarily serves as an auxiliary tool, forming complementarity with human skills.

Continuing based on the s, c data from the table above, we plot the s, c **tier distribution chart**. This chart serves to *reveal GenAI's potential impact pattern on the occupation*: if most tasks fall in the high s low c region, the occupation faces higher automation substitution risk; conversely, if most tasks fall in the low s high c region, GenAI is more likely to augment rather than substitute human work.

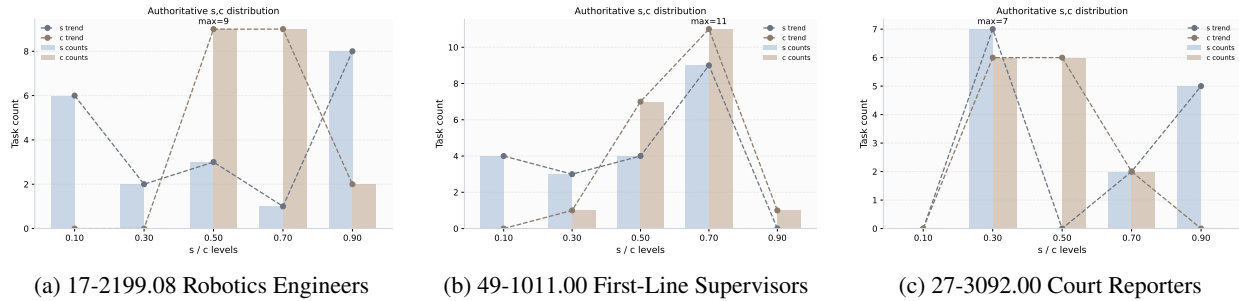


Figure 7: Authoritative s, c tier distribution comparison across three occupations

Combining Figure 7, we explain the impact mechanisms of the three occupation types from the (s_{ij}, c_{ij}) structural distribution, corroborating with the prediction phase results. For **Robotics Engineers**, the focus is on the two high- s and high- c bars on the right side, indicating that GenAI brings substantial net impact, thus requiring greater structural adjustment to adapt. For **First-Line Supervisors (Mechanics/Installation/Repair)**, the rightmost bar is nearly zero, with more tasks evenly distributed across medium-height bars, reflecting task diversity with relatively balanced contribution to overall structural change. For **Court Reporters and Captioners**, the characteristics are very pronounced—the rightmost has only one high- s bar while the corresponding c bar is zero, confirming high dependence on a few core tasks.

After obtaining (s_{ij}, c_{ij}) data, we can calculate μ_{ij}^{task} . $\mu_{ij}^{task} > 0$ indicates GenAI is more likely to augment that task (improving output/quality/speed), while $\mu_{ij}^{task} < 0$ indicates the task is more likely to be substituted or significantly compressed. We can further summarize the weights and typical tasks with negative net impact ($\mu_{ij}^{task} < 0$) across the three occupations, with results in Table 4.

Table 4: Summary of Negative Net Impact Tasks ($\mu_{ij}^{task} < 0$) Across Three Occupations

SOC	Occupation	Count	$\sum w_{ij}$	Typical Tasks (id, μ , dim)
17-2199.08	Robotics Engineers	8	0.400	16521/16516/21057/16528 etc., $\mu = -0.20$, D3
49-1011.00	First-Line Supervisors (Mechanics)	1	0.059	2912, $\mu = -0.20$, D5
27-3092.00	Court Reporters & Captioners	5	0.413	8653/21183/8663/8656/8657, $\mu = -0.40$, D8

- **Robotics Engineers:** Cumulative negative net impact weight is 0.400, with all 8 tasks falling in D3 (Algorithm Programming & Automation), $\mu = -0.20$. This means approximately 40% of core tasks face automation pressure, but the net impact magnitude is relatively moderate ($s = 0.90, c = 0.70$), manifesting as “high substitution but still substantial augmentation” in coding tasks—essentially strong effects canceling out.
- **First-Line Supervisors:** Only 1 task (scheduling and dispatch, task_id=2912) shows negative net impact, with weight only 0.059, falling in D5 (Planning & Workflow Scheduling), $\mu = -0.20$. This occupation’s core tasks mostly fall in supervision, training, and coordination dimensions (D6/D7), forming complementarity with human skills, with the lowest overall substitution risk.
- **Court Reporters & Captioners:** Cumulative negative net impact weight reaches 0.413, with 5 tasks concentrated in D8 (Language Understanding & Text Generation), $\mu = -0.40$, far exceeding the other two occupations. This indicates the occupation’s core tasks (stenography, transcription, proofreading) are highly concentrated in the language automation dimension, with the strongest GenAI substitution effect, expected to show earlier “task share dropping below threshold” time points in subsequent dynamic evolution.

Based on the above analysis, we identify the key driving factors for each occupation:

- **Robotics Engineers:** AI-assisted code generation tools (GitHub Copilot) affect D3 dimension, driving 40% task weight negative impact; GenAI-driven simulation platforms (NVIDIA Isaac [22]) accelerate testing automation while enhancing system-level design efficiency; LLM-robot control integration creates new augmentation space for cross-modal tasks.
- **First-Line Supervisors:** Predictive maintenance AI (IBM Maximo, SAP) affects D5/D7 dimensions [16]; AR/AI diagnostic tools enhance efficiency but core value relies on on-site judgment (D4) and coordination (D6), yielding complementary effects; automated scheduling is the only negative-impact category ($\mu = -0.20$).
- **Court Reporters:** High-accuracy ASR systems (Whisper, WER < 1% [32]) directly impact core transcription tasks—the primary substitution driver; AI captioning services (Zoom, YouTube) reduce market demand; GenAI legal tools (Harvey AI, CoCounsel) automate auxiliary tasks [20].

3.2.2 Coupling “Development Trajectory” with “Task Demand”

The problem requires “given the current trajectory and expected impacts of Gen-AI,” where we define “current trajectory” as an exogenous adoption path $A(t) \in [0, 1]$ representing Gen-AI’s adoption rate at time t . Furthermore, 0 is treated as the current time point, meaning the span from present to future. To ensure this exogenous trajectory’s magnitude is consistent with reality, we reference trend boundaries and scenario upper bounds from organizational-level AI adoption surveys and industry forecasts [17, 18, 6].

Considering model applicability and accuracy, we do not make any quantitative predictions for any future time point, but instead approach from a more fundamental angle, exploring changes in future task relative shares. We introduce several indicators to determine “the essential nature of tasks has changed.”

(Indicator I) Structural Change Contribution Rate First, define the unnormalized task weight

$$q_{ij}(t) = w_{ij} \left[(1 - A(t)) + A(t) \exp(\eta \mu_{ij}^{task}) \right], \quad (8)$$

where $\eta > 0$ is the impact intensity scaling (dimensionless), characterizing “the magnitude of task structure rearrangement brought by AI adoption rate,” with its effects reported in sensitivity analysis.

Intuitively, when $A(t) = 0$, only w_{ij} remains, keeping task weights consistent with current state; as $A(t)$ increases, since the right-side portion applies a multiplicative transformation monotonically consistent with μ_{ij}^{task} (ensuring positive weights), the task-level net impact μ_{ij}^{task} gradually releases, and task demand structure rearranges accordingly.

Then normalizing $q_{ij}(t)$ yields the relative task share.

$$p_{ij}(t) = \frac{q_{ij}(t)}{\sum_{k=1}^{J_i} q_{ik}(t)}. \quad (9)$$

When $A(t)$ rises (GenAI adoption rate increases), if $\mu_{ij}^{task} < 0$, then $q_{ij}(t)$ relatively shrinks and $p_{ij}(t)$ decreases; conversely if $\mu_{ij}^{task} > 0$, that task’s share rises.

For any two time points t_0, t_1 , task j ’s share change is defined as

$$r_{ij}(t_0, t_1) = |\Delta p_{ij}(t_0, t_1)| = p_{ij}(t_1) - p_{ij}(t_0). \quad (10)$$

Sorting by r_{ij} in descending order identifies tasks contributing most to structural change (top contributors).

(Indicator II) Top- m Turnover Rate Define an occupation’s core task set as the top m tasks sorted by $p_{ij}(t)$ in descending order, denoted $\text{Top}_m(p_i(t))$. The turnover rate measures the degree of change in the core task set:

$$J_i(t_0, t_1; m) = 1 - \frac{|\text{Top}_m(p_i(t_0)) \cap \text{Top}_m(p_i(t_1))|}{m}. \quad (11)$$

When $J_i = 0$, the core task set is completely unchanged; when $J_i = 1$, the core task set is completely replaced.

(Indicator III) Rank Inversion and Normalized Inversion Intensity Let $\text{rank}_i(j, t)$ denote task j ’s rank in occupation i sorted by $p_{ij}(t)$ in descending order (1 being highest). Define the inversion count as

$$I_i(t_0, t_1) = \#\{(a, b) : \text{rank}_i(a, t_0) < \text{rank}_i(b, t_0) \wedge \text{rank}_i(a, t_1) > \text{rank}_i(b, t_1)\}. \quad (12)$$

This indicator counts the number of task pairs where “at t_0 , a ranked ahead of b , but at t_1 , a fell behind b .” For cross-occupation comparison, we normalize:

$$\tilde{I}_i(t_0, t_1) = \frac{2I_i(t_0, t_1)}{N_i(N_i - 1)} \in [0, 1], \quad (13)$$

where N_i is the total number of tasks in occupation i . $\tilde{I}_i$, as the normalized form of Kendall τ distance [14], explains the degree of structural change in task relative importance. The above indicators serve as the main criteria for “occupation has been redefined”:

$$\text{Structure Redefined} \iff \tilde{I}_i(t_0, t_1) \geq \theta_I \quad \text{or} \quad J_i(t_0, t_1; m) \geq \theta_J, \quad (14)$$

where $\theta_I \in (0, 1)$ is the inversion intensity threshold (e.g., $\theta_I = 0.15$), $\theta_J \in (0, 1)$ is the turnover rate threshold (e.g., $\theta_J = 0.40$), and m is the number of core tasks (e.g., $m = 5$).

Thresholds θ_I and θ_J are not fixed constants; we compute results under multiple threshold settings and present optimal results.

Based on the three indicators defined above, we can construct two core visualizations: **rank inversion**

diagram (Figure 8) and **contribution decomposition bar chart** (Figure 8 right):

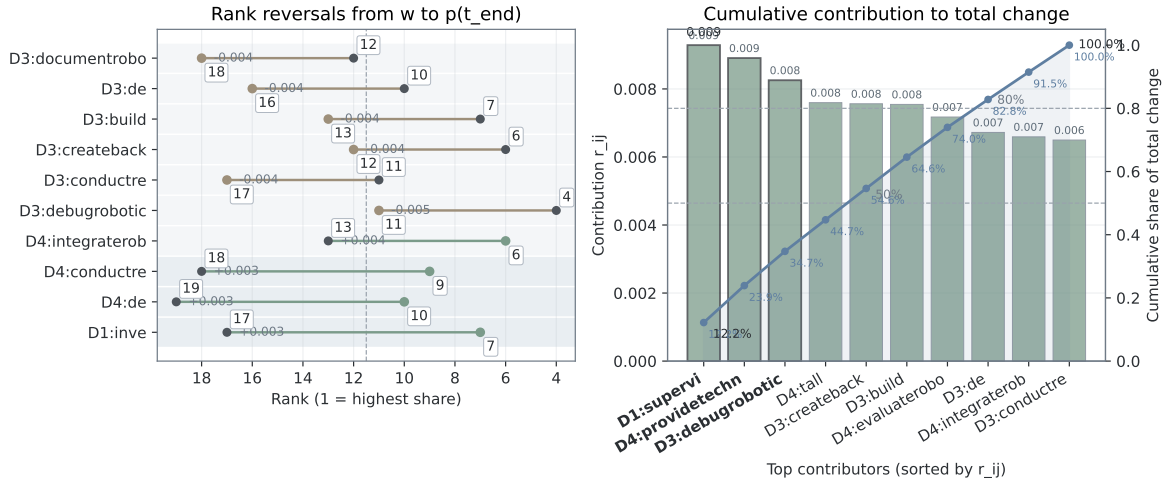


Figure 8: Left: STEM occupation rank inversions. Line endpoints represent $\text{rank}_i(j, t_0)$ and $\text{rank}_i(j, t_1)$; larger inversion magnitude indicates more significant structural rearrangement. Right: Top contributor decomposition of STEM occupation structural change. Tasks shown in descending r_{ij} order with cumulative contribution to total share change, revealing the main driving tasks for structural rearrangement.

The main text only shows visualizations for Robotics Engineers; the other two occupations use the same modeling approach to obtain data results, with detailed analysis in Section 3.2.3.

3.2.3 Exploring the Future of the Three Selected Occupations

Based on the task prediction model, GenAI capability mapping (Table 3), and the structural change indicators defined above, we now comprehensively analyze results and explore each occupation’s future trajectory under the GenAI adoption path.

Our analysis indicates that Robotics Engineers exhibit a balanced but augmentation-leaning impact profile. Figure 7a shows complementarity scores c concentrated in mid-to-high tiers, while approximately 40% of task weight falls in the negative net impact region, primarily concentrated in dimension D3 (Algorithm Programming & Automation), with $\mu = -0.20$. However, this portion of substitution pressure is offset by strong augmentation effects in engineering problem-solving (D1) and data analysis (D2). This “give-and-take” symmetric pattern indicates: GenAI is not simply “eliminating” certain task types, but triggering task weight rebalancing within the occupation.

Under the baseline scenario with $\eta = 1.0$, reconfiguration mass grows from $R(t_0) = 0.086$ to $R(t_{\text{end}}) = 0.121$, indicating approximately 12% cumulative structural transformation when adoption rate saturates at $A = 0.95$. Top-5 turnover rate stabilizes at $J_5 = 0.40$, meaning two of the five most important tasks are replaced by previously secondary tasks. As shown in Figure 8, rank inversion patterns indicate declining relative shares for routine coding and documentation tasks, while system-level integration, hardware-software co-design, and cross-functional collaboration tasks rise.

We conclude that future robotics engineers will primarily be responsible for orchestrating AI-assisted development workflows, validating machine-generated solutions, and focusing on tasks requiring physical-world judgment and creative problem-solving—capabilities current GenAI cannot replicate.

For First-Line Supervisors of Mechanics, Installation, and Repair, this occupation exhibits the lowest substitution risk among the three. The (s, c) distribution (Figure 7b) is notably dispersed, with no single dimension dominating the impact profile. Only one task (scheduling and dispatch, task_id = 2912) has negative net impact, contributing only 0.059 of total task weight. Dominant dimensions—D4 (On-Site Perception & Physical Oper-

ation), D6 (Collaboration & Training), and D7 (Supervision & Quality Control)—all exhibit positive or neutral ρ_k values, indicating GenAI’s complementary rather than substitutive effects.

Under the baseline scenario ($\eta = 1.0$), reconfiguration mass grows modestly from $R(t_0) = 0.061$ to $R(t_{\text{end}}) = 0.086$, the smallest structural change among the three occupations. However, Top-5 turnover rate reaches $J_5 = 0.80$, which appears contradictory at first glance. This high turnover rate reflects the flat weight distribution characteristic of supervisory roles: many similarly important tasks compete for Top-5 positions, making rankings sensitive to minor perturbations even with limited total reconfiguration.

Future first-line supervisors will remain structurally stable but operationally enhanced. GenAI tools will enhance diagnostic capabilities, streamline scheduling processes, and improve training delivery, but the core value proposition—real-time judgment in unpredictable physical environments, interpersonal coordination, and hands-on troubleshooting—remains firmly human-anchored. This occupation embodies the “AI as copilot” paradigm rather than transformation or substitution.

Court Reporters and Simultaneous Captioners (27-3092.00): Substitution-Dominant Disruption Court reporters face the most severe substitution pressure among the three occupations. The (s, c) distribution (Figure 7c) shows high-weight tasks clearly concentrated in dimension D8 (Language Understanding & Text Generation), where $s = 0.90$ and $c = 0.50$ produce strong negative net impact $\mu = -0.40$. The cumulative weight of negative-impact tasks reaches $\sum_{j: \mu_{ij} < 0} w_{ij} = 0.413$, the highest among the three occupations.

Under the baseline scenario ($\eta = 1.0$), reconfiguration mass grows from $R(t_0) = 0.060$ to $R(t_{\text{end}}) = 0.092$. While the absolute magnitude appears moderate, the direction of change is unambiguous: core transcription and drafting task shares decline rapidly, while the relative importance of quality verification, legal context interpretation, and certification tasks rises. The $J_5 = 0.60$ Top-5 turnover rate confirms substantial restructuring of the task hierarchy.

Future court reporters will experience fundamental role redefinition. As ASR and GenAI transcription tools achieve near-human accuracy in controlled acoustic environments, the occupation shifts from stenographic production toward quality assurance and legal certification. Practitioners will increasingly serve as expert validators, ensuring transcription accuracy in adversarial legal environments, interpreting domain-specific terminology, and maintaining chain of custody for official records. Those who fail to acquire complementary skills in legal technology, quality management, or specialized domains will face significant substitution risk.

3.3 Educational Programs and Gen-AI Integration Strategies

Based on the aforementioned occupation selection and task-level Gen-AI impact analysis, we recommend one representative educational program for each of the three occupations. We follow three criteria: **whether the institution demonstrates strong industry leadership in the relevant sector, whether the curriculum aligns well with industry demands, and whether the institution already possesses Gen-AI-related resources.** Specific selections and rationale are provided in Appendix Table 10.

For **Robotics Engineers**, we recommend Carnegie Mellon University’s Robotics Institute MRSD program, which possesses industry leadership and has established Gen-AI-related curriculum resources. For **First-Line Supervisors of Mechanics**, Universal Technical Institute (UTI) demonstrates the ability to rapidly respond to industry technology transitions and has already incorporated AI diagnostics into its educational narrative. For **Court Reporters**, Generations College, as the historical inheritor of America’s first court reporting program, offers a Realtime training system that complements Gen-AI era editing tools.

3.3.1 Meta-Model: Employment–Humanistic–AI Overlap

After determining the recommended programs, we construct a **curriculum-level evaluation framework** to guide course reform recommendations within each program. The model’s core idea is: each course is evaluated along three orthogonal dimensions—*employability*, *humanistic value*, and *AI overlap*—thereby yielding three directional choices: “Keep / Transform / Prune.” The employment axis focuses more on labor market demands; the humanistic axis is primarily subjectively defined; the AI overlap axis focuses solely on the connection between course content and Gen-AI capability dimensions.

Let the educational program contain M courses, with course index $m = 1, \dots, M$, and course text description

t_m (course title and description). The target occupation set is denoted O , with occupation $o \in O$ corresponding to the O*NET task database. Define:

- w_o : Labor market demand weight for occupation o (set to uniform distribution if unavailable);
- $v_{o,j}$: Saliency weight of task j in occupation o , obtained from O*NET importance×frequency normalization; this is the method already implemented in Question 2.
- $s_{m,o,j} = \text{sim}(t_m, \text{task}_{o,j})$: Semantic coverage of course m for task (o, j) (based on text similarity);
- b_k : Humanistic dimension anchor text, $k \in \{\text{civic, agency, wellbeing, lifelong}\}$;
- $k = 1, \dots, 9$: The aforementioned nine-dimensional Gen-AI capability dimension index;
- $w_{m,k} \in [0, 1]$: Normalized footprint of course m in dimension k , $\sum_{k=1}^9 w_{m,k} = 1$ (extracted from course title and description);
- (M_k, s_k, c_k, ρ_k) : Nine-dimensional capability scale parameters, where $\rho_k = c_k - s_k$.

(Axis I) **Employability** The core question of the employment axis is: what is the course's *marginal contribution* to students entering the target occupation set O ? We use “occupation-demand-weighted task coverage rate” to characterize the program's match with the labor market.

First, define the program's coverage of task (o, j) as the maximum similarity across all courses:

$$c_{o,j} = \max_{m \in \{1, \dots, M\}} s_{m,o,j}. \quad (15)$$

The overall employability score is:

$$E = \sum_{o \in O} w_o \cdot \sum_{j \in J(o)} v_{o,j} \cdot c_{o,j}. \quad (16)$$

To measure a single course's contribution, we define **marginal employment value**. Let $E^{(-m)}$ denote the employability score after removing course m :

$$E^{(-m)} = \sum_{o \in O} w_o \cdot \sum_{j \in J(o)} v_{o,j} \cdot \max_{n \neq m} s_{n,o,j}. \quad (17)$$

Then course m 's marginal contribution is:

$$\Delta E_m = E - E^{(-m)}. \quad (18)$$

The interpretable meaning of ΔE_m is: if course m is removed, how much will the program's alignment with the current labor market decrease; the larger ΔE_m , the more *irreplaceable* the course is on the employment axis.

(Axis II) **Humanistic Value** The humanistic axis focuses on a course's cultivation of students' *non-occupational capabilities*. We use anchor text similarity as a concise and reproducible metric:

$$H_m = \sum_{k \in \{\text{civic, agency, wellbeing, lifelong}\}} \text{sim}(t_m, b_k). \quad (19)$$

where b_k are predefined humanistic anchor texts. The four dimensions and their anchor keywords are:

- **Civic Literacy**: ethics, public, civic, law, compliance, responsibility;
- **Agency**: decision, leadership, management, planning, strategy, autonomy;
- **Well-being**: safety, health, wellbeing, risk, care, quality;
- **Lifelong Learning**: learning, training, skill, development, professional, continuous.

This axis identifies courses that, while not directly improving short-term employment matching, contribute to students' long-term development and social public value.

(Axis III) AI Overlap The AI overlap axis measures the *substitution pressure* between course content and Gen-AI capability dimensions. Unlike approaches based on keywords or opaque dictionaries, we define the AI-Overlap metric as the weighted overlap between “course nine-dimensional capability footprint” and “Gen-AI’s independent delivery strength in each dimension,” fully reusing the aforementioned nine-dimensional capability scale parameters (M_k, s_k, c_k, ρ_k) , thereby avoiding subjective weighting.

First, define dimension gating (two calibers):

The only difference lies in the domain; using 1 to avoid affecting numerical values.

$$I_k = \mathbf{1}[\rho_k \leq 0], \quad I_k^{\text{strict}} = \mathbf{1}[\rho_k < 0]. \quad (20)$$

where I_k allows “uncertain/balanced” dimensions to enter the pressure measure, while I_k^{strict} only counts “explicitly substitution-oriented” dimensions into pressure, used to construct conservative pruning conditions. Define dimension-level substitutability intensity as:

$$a_k = M_k \cdot s_k \cdot I_k, \quad a_k^{\text{strict}} = M_k \cdot s_k \cdot I_k^{\text{strict}}. \quad (21)$$

Then course m ’s AI-Overlap (two calibers) is:

$$A_m = \sum_{k=1}^9 w_{m,k} a_k, \quad A_m^{\text{strict}} = \sum_{k=1}^9 w_{m,k} a_k^{\text{strict}}. \quad (22)$$

A_m reflects the overall overlap pressure between the course and Gen-AI capability dimensions, while A_m^{strict} reflects the conservative pressure from “proven substitution” dimensions. This definition maintains consistency with the capability dimension research in the previous two questions and depends only on course text (title and description).

Three-Axis Decision Rules: Keep / Transform / Prune Based on the three-axis metrics, we provide course decision rules. The intuitive logic is as follows:

- **Prune:** The course simultaneously satisfies “low employment margin,” “low humanistic value,” and “extremely high strict AI overlap.” Such courses lack both employment and humanistic support, and their core outputs are highly deliverable independently by Gen-AI; we recommend removing them from the curriculum.
- **Transform:** The course has high AI overlap but does not satisfy the conservative pruning conditions. Such courses should not be deleted but redesigned (e.g., compressing outputs that can be directly generated by Gen-AI, shifting toward human-led process evidence, on-site reasoning, ethical deliberation, and human-AI collaboration capabilities).
- **Keep:** Remaining courses maintain their current form.

To ensure cross-institutional portability and avoid fabricating constants, we define thresholds as empirical quantiles within the program (using credit-weighted quantiles when course credits are available; otherwise, uniform quantiles). Let

$$e_{\text{low}} = Q_{0.25}(\Delta E_m), \quad h_{\text{low}} = Q_{0.25}(H_m), \quad a_{\text{high}} = Q_{0.75}(A_m), \quad a_{\text{strict_high}} = Q_{0.90}(A_m^{\text{strict}}). \quad (23)$$

The final decision rules are:

$$\text{PRUNE}(m) = \mathbf{1}\left[(\Delta E_m \leq e_{\text{low}}) \wedge (H_m \leq h_{\text{low}}) \wedge (A_m^{\text{strict}} \geq a_{\text{strict_high}})\right], \quad (24)$$

$$\text{TRANSFORM}(m) = \mathbf{1}\left[\neg \text{PRUNE}(m) \wedge (A_m \geq a_{\text{high}})\right], \quad (25)$$

$$\text{KEEP}(m) = 1 - \text{PRUNE}(m) - \text{TRANSFORM}(m). \quad (26)$$

We apply the above three-axis decision rules to the three representative educational programs, with results summarized in Table 5.

Table 5: Course-Level Transformation Recommendations Across Three Programs. All flagged courses receive TRANSFORM labels; no courses qualify for PRUNE, reflecting the model’s conservative stance toward course elimination. A^s denotes A^{strict} .

Program	Course (Code)	ΔE	H	A	A^s	Top-2 Dims
<i>Generations Court Reporting</i> (1 course flagged for transformation)						
	CR-CAT-1 Machine Shorthand Theory	1.000	0.0	0.584	0.453	D8=0.53, D9=0.42
<i>MRSD, Carnegie Mellon University</i> (3 courses flagged for transformation)						
	16-642 Manipulation, Estimation & Control	0.000	0.0	0.330	0.330	D3=0.53, D4=0.45
	16-697 Intro to Robotics Business	0.000	0.5	0.371	0.371	D3=0.51, D4=0.33
	16-698 Adv Topics in Robotics Business	0.000	0.0	0.404	0.404	D3=0.64, D4=0.31
<i>UTI Diesel Technology</i> (2 courses flagged for transformation)						
	DT13-162 Steering & Suspension Systems	0.000	0.0	0.336	0.058	D7=0.48, D2=0.18
	DT13-211 Diesel Engines	0.000	0.0	0.380	0.055	D7=0.60, D2=0.12

3.3.2 Institution-Level Strategic Recommendations

Based on the results from the above meta-model, it is noteworthy that the prune lists for all three programs are empty, with all recommendations falling under “Transform” rather than “Prune.” This result aligns well with reality: educational reform proceeds incrementally rather than through unrealistic wholesale elimination. We further provide targeted Gen-AI integration strategy recommendations for the leaders of the three institutions.

Generations College: Court Reporting Program

- i. **Redefine core competencies: From “speed” to “judgment”.** Traditional court reporting emphasizes real-time transcription speeds above 225 words per minute, while Gen-AI’s advancing speech recognition capabilities are changing the boundaries of “transferable skills” that students need to train. We recommend shifting the curriculum focus from “mechanical speed training” to “real-time verification and semantic judgment”—cultivating students’ ability to perform legal terminology correction, speaker identification, and courtroom context understanding on AI drafts [19].
- ii. **Introduce human-AI collaboration workflows.** The curriculum should design AI-assisted stenography experimental modules, allowing students to perform human intervention on AI real-time transcription outputs, annotate uncertain passages, and conduct attribution analysis of AI errors. This not only enhances students’ awareness of AI limitations but also lays the foundation for human-AI collaboration in their future careers.
- iii. **Strengthen legal ethics and responsibility assessment.** Court records have legal effect, and responsibility for AI-generated content accuracy ultimately rests with humans. The curriculum should add an “AI output auditing” unit to cultivate students’ ability to identify AI hallucination, bias, and privacy leakage risks.

Carnegie Mellon University: MRSD Program

- i. **From “algorithm implementation” to “system integration and boundary conditions”.** Gen-AI can already generate high-quality control algorithm code and optimization solvers. We recommend that 16-642 *Manipulation, Estimation & Control* compress traditional algorithm derivation and coding training, shifting emphasis toward: (a) sensor noise, actuator delay, and environmental uncertainty in real physical systems; (b) safety auditing and boundary condition testing of AI-generated code; (c) fault recovery and ethical decision-making in human-AI collaboration scenarios.

- ii. **Business courses focus on “non-automatable” strategic judgment.** The two business courses 16-697 and 16-698 should reduce dependence on content that Gen-AI can rapidly generate, such as market analysis reports and financial models, shifting emphasis toward: (a) strategic selection and timing judgment of technical roadmaps; (b) cross-cultural team management and stakeholder negotiation; (c) case studies on technology ethics and social responsibility.
- iii. **Establish “AI literacy” horizontal modules.** Given that MRSD students will lead robotics R&D teams in the future, we recommend introducing AI literacy required modules at the program level, covering: capability boundaries of large language models, prompt engineering best practices, and explainability and auditing methods for AI systems; this aligns with higher education evidence on “elevating AI capabilities from individual experimentation to institutionalized curriculum design and policy frameworks” [13, 37].

Universal Technical Institute: Diesel Technology Program

- i. **Incorporate AI diagnostic tools into hands-on training.** Modern diesel vehicles are equipped with OBD-II diagnostic interfaces and AI fault prediction systems. We recommend adding an “AI diagnostic tools hands-on” module to the curriculum, allowing students to learn how to interpret AI-generated fault codes, evaluate the reliability of AI recommendations, and make physical inspection and repair decisions based on AI prompts.
- ii. **Strengthen physical skills that “AI cannot replace”.** Although AI can assist diagnosis, actual disassembly, calibration, and repair still require human technicians. The curriculum should clearly distinguish between “AI-assistable” and “AI-irreplaceable” skill boundaries, and increase hands-on practical weighting in assessments while reducing dependence on theoretical knowledge memorization.
- iii. **Introduce electrification and hybrid powertrain transition content.** As the transportation industry transitions toward electrification, market demand for traditional diesel technology will gradually shrink. We recommend adding electric and hybrid system modules to the curriculum, helping students build adaptability across technology platforms and reducing career risk from single technology paths.

Cross-Institutional Common Recommendations Beyond the institution-specific recommendations above, we propose three cross-institutional common principles:

- **Assessment system transformation: From “output” to “process”.** Traditional assessment focuses on students’ final submitted work (papers, code, reports), while Gen-AI makes “automatic generation” of these outputs possible. We recommend shifting assessment focus toward process evidence—such as live oral defenses, real-time problem-solving, and collaboration process documentation—to ensure assessment validity and fairness [11].
- **Faculty development first.** The prerequisite for curriculum transformation is faculty possessing Gen-AI literacy. Institutions should prioritize investment in faculty training, enabling them to understand Gen-AI capability boundaries, master human-AI collaborative instructional design methods, and demonstrate responsible AI use in the classroom [13].
- **Continuous monitoring and iteration.** Gen-AI capabilities are rapidly evolving, and currently “irreplaceable” dimensions may be breached in the future. We recommend institutions establish annual curriculum audit mechanisms, periodically re-evaluate three-axis metrics, and dynamically adjust curriculum design according to Gen-AI capability evolution; meanwhile, clear institutionalized policies on “allowing use but setting boundaries” are needed to reduce the learning and integrity tensions from gray areas [10, 12].

4 Simulation Results and Sensitivity Analysis

4.1 Sensitivity Analysis of Occupation Selection

In the maximum-area triangle search, exhaustive enumeration of all occupation combinations has computational complexity $O(n^3)$, where n is the total number of occupations. When the O*NET database contains over 1,000 occupations, this brute-force enumeration is too time-consuming and resource-intensive. To address this, we adopt a “candidate pruning” strategy: for each domain (STEM/Trade/Arts), we select the top M occupations farthest from the domain center, plus the most extreme occupations on each feature axis, forming a candidate pool. We only perform the triple loop search within this candidate pool.

Validation Method To verify that pruning does not miss the true optimal solution, we perform a sensitivity scan on candidate pool size M . Specifically, we take $M \in \{200, 400, 800\}$ across three tiers:

- $M = 200$: Compact candidate pool with highest computational efficiency;
- $M = 400$: Candidate pool expanded by a factor of two, covering more edge-case occupations;
- $M = 800$: Candidate pool approaches full domain occupation size, approximating complete enumeration.

The solution results for the maximum-area triangle under the three candidate pool sizes are shown in Table 6.

Table 6: Candidate Pool Size Sensitivity Test: Optimal Triangle Solutions Under Different M

Candidate Size M	Triangle Area	Shortest Edge	Three Points Consistent	Area Difference
200	0.2622	0.7514	Baseline	—
400	0.2622	0.7514	Consistent	0%
800	0.2622	0.7514	Consistent	0%

All three solutions return the same occupation combination/plane size. The same “occupational triangle” consistently wins across all three tests, proving the result is robust (reliable) rather than a mathematical coincidence. In this test, the area difference is 0%, far below the 5% threshold standard; therefore, we determine that the three-occupation configuration is robust to candidate pool size selection.

Each output yields the identical “occupational triangle,” and in this test, the area difference is 0%, far below the 5% threshold standard; therefore, we determine the model is robust rather than a mathematical coincidence.

4.2 Structural Sensitivity of Employment Destination O

The employment axis’s sole exogenous semantic input is the target occupation set O . If O ’s definition scope changes, $J(o)$, $v_{o,j}$, and ΔE_m will undergo systematic shifts, thereby altering the meaning of “labor market match rate.” To test the model’s cross-institutional extrapolation boundaries, we vary only O , fixing H_m , A_m , A_m^{strict} , and the quantile threshold strategy, with w_o set to uniform distribution.

We construct three scope tiers based on O*NET’s Related Occupations (Primary-Short/Primary-Long): O_{narrow} as the most direct SOC, O_{medium} adding 1–2 neighboring SOCs, and O_{wide} further expanding by 2–4 related SOCs. The three-tier O definitions are shown in Table 7.

Table 7: Three-Tier Scope of Target Occupation Set O (SOC)

Program	$O_{\text{narrow}} / O_{\text{medium}} / O_{\text{wide}}$
MRSD (CMU)	17-2199.08 17-2199.08, 17-2199.05, 17-2072.00 17-2199.08, 17-2199.05, 17-2072.00, 17-2112.03, 17-2141.02, 17-2011.00, 17-2071.00
UTI Diesel	49-1011.00 49-1011.00, 51-1011.00, 47-1011.00 49-1011.00, 51-1011.00, 47-1011.00, 43-1011.00, 37-1011.00, 39-1014.00, 45-1011.00
Generations Court Reporting	27-3092.00 27-3092.00, 43-6012.00, 43-4031.00 27-3092.00, 43-6012.00, 43-4031.00, 43-4021.00, 31-9094.00, 43-9061.00, 43-9021.00

For each $O^{(r)}$, we recalculate $E^{(r)}$ and $\Delta E_m^{(r)}$, recalculating only the employment threshold $e_{\text{low}}^{(r)}$ while keeping the three-axis rules unchanged. Robustness results are shown in Table 8. The TRANSFORM set consistency across all three programs under all three scope tiers is 1.00, and all course label stability rates are 1.00. Meanwhile, the ranking stability of ΔE significantly decreases for UTI and Court Reporting ($\tau = 0.29/0.14$), indicating that the employment axis is sensitive to O ’s definition scope.

Table 8: Robustness Metrics Under O Definition Scope Variation (Narrow vs. Wide)

Program	Transform Jaccard J	Kendall τ	Min Course Stability Rate
MRSD (CMU)	1.00	0.83	1.00
UTI Diesel	1.00	0.29	1.00
Generations Court Reporting	1.00	0.14	1.00

Based on this, we establish extrapolation boundaries: when J is high, core reform recommendations are primarily driven by AI-overlap, with strong cross-institutional generalizability; when τ significantly decreases, the employment axis’s dependence on O increases, and extrapolation requires re-specifying O ’s definition scope. We therefore emphasize: **The model is transferable, but O cannot be transferred by default.**

5 Discussion: Strengths and Limitations

5.1 Strengths

First Principles: Occupations Are Composed of Tasks Our modeling approach stems from a simple first principle—the **essence of an occupation is its task composition**. Since GenAI impacts specific tasks rather than abstract “occupation labels,” studying occupational evolution should start from the task level. Existing empirical evidence also indicates that GenAI adoption is reshaping work content and skill demand structures, and this change first manifests in the task and skill composition within positions, rather than simply equating to “a certain occupation being substituted/not substituted” [9]. Based on this, we propose the concept of **Task DNA**, decomposing each occupation into a measurable task weight vector, making the vague question of “AI’s impact on occupations” computable and traceable.

Innovation: From “Is This Occupation at Risk?” to “How Are Tasks Being Rearranged?” Traditional research often provides vague conclusions like “Occupation A is high-risk / Occupation B is safe.” Our model answers more granular questions: which task shares will contract, which will expand, and when occupational boundaries trigger redefinition. This task-level granularity of analysis aligns with research conclusions in education and labor studies that “career readiness needs to be defined by human-machine complementary skills rather than static job labels,” thus having greater interpretability and policy guidance value [19, 37].

5.2 Limitations

Due to the short development period of Gen-AI, we lack sufficient time-series data to demonstrate intuitive structural mutations. Furthermore, as shown in Figure 9, the magnitude of change in the structural curves is very small, resulting in effects that are not visually pronounced.

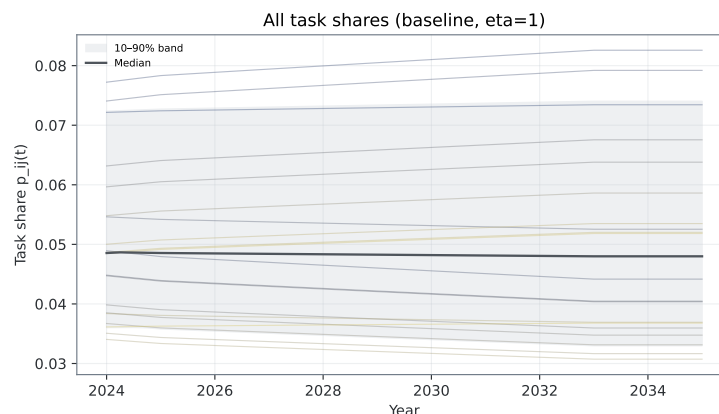


Figure 9: Evolution of all task shares over time for the STEM occupation under the baseline scenario.

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Appendices

.1 O*NET Descriptor Access Links

Table 9 provides the access links and descriptor IDs for the six O*NET descriptors used in the three-dimensional feature mapping.

Table 9: O*NET Descriptor Access Links and Descriptor IDs (for Reproducibility)

Feature Axis	Descriptor Name	Descriptor ID	URL
$digit(x)$	Working with Computers	4.A.3.b.1	https://www.onetonline.org/find/descriptor/result/4.A.3.b.1
	Electronic Mail	4.C.1.a.2.h	https://www.onetonline.org/find/descriptor/result/4.C.1.a.2.h
$physical(x)$	Using Hands to Handle Objects	4.C.2.d.1.g	https://www.onetonline.org/find/descriptor/result/4.C.2.d.1.g
	Kneeling/Crouching/Stooping	4.C.2.d.1.e	https://www.onetonline.org/find/descriptor/result/4.C.2.d.1.e
$iprisk(x)$	Originality	1.A.1.b.2	https://www.onetonline.org/find/descriptor/result/1.A.1.b.2
	Thinking Creatively	4.A.2.b.2	https://www.onetonline.org/find/descriptor/result/4.A.2.b.2

.2 Two-Dimensional Projection Area Formula

To ensure the heterogeneity measure under two-dimensional projection is consistent with the three-dimensional definition, we define the two-dimensional triangle area for any projection π (e.g., $(digit, physical)$, $(digit, iprisk)$, or $(physical, iprisk)$) as

$$\text{Area2D}_{\pi}(x_S, x_T, x_A) = \frac{1}{2} |(\pi(f(x_T)) - \pi(f(x_S))) \times (\pi(f(x_A)) - \pi(f(x_S)))|,$$

where $\times$ denotes the cross product (scalar) operation of two-dimensional vectors, and $\pi(\cdot)$ is the projection mapping from three-dimensional features to two-dimensional coordinates.

.3 Supplementary Note: Task DNA Text Legends

Due to page limitations, we did not list all task texts for the three occupations in the main text. When generating Figure 5, we simultaneously exported weight-sorted task text legends for reproducibility and auditing purposes.

.4 Educational Program Recommendation Details

Table 10 summarizes the recommended educational programs and selection rationale for the three representative occupations.

Table 10: Recommended Educational Programs and Selection Rationale

Occupation & Institution Type	Recommended Institution & Program	Selection Rationale
Robotics Engineers (University)	Carnegie Mellon University Robotics Institute: Master of Science in Robotic Systems Development (MRSD)	(1) Institution-level “pathfinder” attributes: Since its establishment in 1979, the Robotics Institute has emphasized its leading position in robotics research and education [3]. (2) Program objectives align with industry and systems engineering capabilities: MRSD is positioned as an advanced industry-practice-oriented degree, emphasizing multidisciplinary technical and business capabilities, and includes internship pathways related to industry partners [2]. (3) Gen-AI-compatible on-campus resources are explicitly available: The university offers a “Generative AI and Large Models” graduate certificate program, which can serve as an institutionalized resource interface for introducing LLM/Gen-AI (e.g., natural language planning, human-robot collaboration, generative design) into robotics courses [1].
First-Line Supervisors of Mechanics, Installers, and Repairers (Vocational/Technical School)	Universal Technical Institute (UTI) Automotive and Diesel Technology Programs	(1) Strong industry alignment and curriculum iteration capability: UTI developed new BEV (Battery Electric Vehicle) courseware through collaboration with Bosch as part of its EV training strategy, demonstrating its ability to rapidly respond to industry technology transitions [35]. (2) Realistic foundation for incorporating “AI-enhanced diagnostics/damage estimation” into skill contexts: UTI’s public content already directly discusses “AI-supported diagnostics” for identifying hidden damage and assisting in estimating repair parts and labor hours, indicating entry points for incorporating AI tool chains in educational narratives and training modules [36].
Court Reporters and Simultaneous Captioners (Arts/Specialized School)	Generations College (formerly MacCormac School) Court Reporting Associate Degree (A.A.S.)	(1) Strong program history and industry representation: The school explicitly claims it established the “first” court reporting program of its kind in the United States in 1912, and emphasizes its industry reputation on enrollment pages [7]. (2) Realtime training is explicitly required in program objectives: The online program page explicitly states “gain Realtime reporting experience in core courses” [8]. (3) Connection points with industry quality standards exist: The NCRA (National Court Reporters Association) published “approved programs” list includes Generations College (with approval validity period noted), serving as external endorsement evidence for program quality and realtime education commitment [21].

Report on Use of AI

OpenAI ChatGPT (Feb 2026, GPT-5.2 Thinking). We used AI as a *research assistant and technical editor* to accelerate model design, verify internal logical consistency, and polish exposition. All modeling choices, assumptions, and final claims were determined and validated by the team.

1. Decision-rule simplification under data availability constraints

Query: Our initial curriculum model is too complex and requires unavailable fields (e.g., credits). Can we redesign it to depend only on course title/description while remaining defensible?

Output: AI recommended (i) removing course-to-course redundancy to avoid unnecessary data requirements, (ii) using only text-derived course representations, and (iii) defining thresholds endogenously via within-program empirical quantiles (credit-weighted if available, uniform otherwise). This kept the rule set simple, portable, and reproducible while preserving interpretability.

2. Data source identification and cross-validation

Query: Where can we find reliable and authoritative public data sources relevant to this study, and how should we cross-validate them?

Output: AI recommended authoritative sources including O*NET OnLine for occupational task structures, Bureau of Labor Statistics (BLS) for employment projections, and peer-reviewed literature for GenAI capability assessments. For cross-validation, AI suggested triangulating task importance scores across multiple O*NET releases and comparing employment trends with independent forecasts.

3. Robustness design and interpretation (occupation-set O perturbations)

Query: The prompt asks whether our framework generalizes across institutions/programs. What is the single most critical sensitivity analysis, and how should we report it?

Output: AI identified the target occupation set O as the dominant external input and designed a narrow/medium/wide O perturbation test. AI also prescribed stability metrics (label stability, TRANSFORM-set Jaccard, employability-rank stability) and helped interpret results conservatively: policy labels remained stable under reasonable O perturbations, while fine-grained employability ranking can vary—therefore the model is transferable, but O must be re-specified when extrapolating to new contexts.

Human verification. We verified all intermediate outputs (course labels, threshold values, and robustness summaries) for logical consistency with the final decision rules, and revised any inconsistent intermediate drafts before inclusion in the paper.